Urbanicity and Fertility: Evidence from Refugees to Canada*

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Abstract

Fertility rates are significantly lower in urban areas than in rural areas. Direct evidence on the nature of this relationship remains limited. Using data from the Canadian Longitudinal Immigration Database, I exploit the exogenous placement of refugees across space to decompose the direct (causal) and indirect (compositional) effects of urbanicity on fertility. Using a two-stage least squares approach, I find that rural-urban fertility differences persist after controlling for selection into cities, suggesting that causal factors are primarily responsible for differences in birth rates. These results have major implications for future population growth as urbanicity rates increase across the world.

Keywords: fertility, refugees, urbanicity, rural-urban differences, Canada

JEL Codes: J13, R2, J11, J15

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1 Introduction

Global birth rates are decreasing, raising concerns among economists and policymakers about a potential future with non-positive population growth (e.g. Jones, 2022). Concurrent with these demographic shifts in fertility, a larger fraction of the global population now resides in cities. It is well-established that urban areas have lower levels of fertility than rural areas. These rural-urban differences in fertility have been documented for at least 70 years, highlighting both their prevalence and persistence across time (United Nations, 1953).

Despite the ubiquity of these differences, it remains unclear to what extent the relationship between urbanicity and fertility is direct (driven by causal factors related to the urban environment) or indirect (driven by differences in who chooses to live where). While some compositional factors, such as education, race, and ethnicity, can be accounted for in cross-sectional analysis, the causal relationship is especially difficult to isolate in the presence of latent differences such as preferences for fertility. This paper provides novel estimates of the direct, causal effect of urbanicity on fertility by exploiting the exogenous placement of refugees throughout Canada.

Using data from the Longitudinal Immigration Database (IMDB) in Canada, I implement a two-stage least squares (2SLS) specification using destination urbanicity as an instrument for the current urbanicity of a refugee to Canada. Destinations are exogenously assigned to potential refugees by Immigration, Refugees, and Citizenship Canada (IRCC). While assignment may consider observable characteristics (e.g. language, country of origin), these refugee placements crucially do not consider latent preferences for urbanicity or fertility. I compare these rural-urban differences to OLS estimates to isolate the causal impact of urbanicity on fertility.

This paper documents three key findings. First, refugee allocations do not represent their preferences for urbanicity. Refugees are more likely to have a initial destination in Canada other than the three largest cities (Toronto, Montreal, and Vancouver) than non-refugee immigrants (NRI's). Once in Canada, rural-allocated refugees are much more likely to move to a large city than non-refugee immigrants. This suggests mismatch between refugee preferences and initial allocations, highlighting the exogeneity of refugee placements in Canada.

Second, selection based on observable characteristics contributes, at most, moderately to differences in birth rates across urbanicities. The inclusion of fixed effects for age, education, family size, place of birth, year, and landing year does not qualitatively affect the estimated effect of urbanicity on fertility. Of the included fixed effects, only accounting for landing year even quantitatively affects the estimated coefficients. This may suggest that the allocation process has changed over time, possibly taking into account different observable characteristics of refugees upon arrival. The fact that flexible controls for these various factors affect estimated coefficients in a limited manner suggests that differences in the composition of the rural and urban populations only moderately contribute to the observed rural-urban fertility gap.

Lastly, urban allocations of refugees cause lower fertility rates among that population. Estimated coefficients representing the effect of urbanicity on fertility are negative and statistically significant. This is true not only for the naive OLS regression but also using a refugee's initial allocation as either a proxy or an instrument for current urbanicity. The estimated causal effect of urbanicity ranges from 0.48 (Proxy) to 0.85 fewer births per 100 women per year (2SLS). This represents between 5.1% and 9.1% of the average birth rate for refugee women in rural areas and is not statistically different from the rural-urban fertility gap for non-refugee immigrants (0.92 births per 100 women per year, or 10.7%). This suggests that causal factors, rather than compositional factors, are the primary driver behind rural-urban fertility differences.

Rural-urban differences in fertility have been documented since at least United Nations (1953), but the literature on the causal nature of this relationship finds mixed results. Kuznets (1974) notes that migration from rural to urban areas is not correlated with changes in birth rates, suggesting that differences are primarily driven by selection.[1] On the other hand, recent research using longitudinal microdata from Finland and the United Kingdom argues that contextual factors play some role in rural-urban fertility differences (Kulu, 2013; Kulu and Washbrook, 2014). Lerch (2018) emphasizes the role of rural-to-urban migration in mitigating these rural-urban differences in the presence of declining urban fertility. The present paper contributes to the existing literature on the relationship between urbanicity and fertility by exploiting the exogenous placement of refugees à la Card (1990) and Foged and Peri (2016). While other papers have utilized cross-country differences or longitudinal household

[1]If cities directly cause lower fertility rates, then cross-country differences in the reallocation of people from rural to urban areas would presumably lead to different changes in aggregate birth rates.

microdata, the results discussed here represent the first quasi-experimental evidence for the causal impact of urbanicity on fertility.

These results have important implications for future population growth. As the world becomes increasingly urbanized, one might expect populations to grow more slowly than in the past. The impact of this change on economic growth depends on the specific context; high population growth has long been considered an obstacle to economic growth in developing countries (Solow, 1956; Kelley and Schmidt, 1995, 2005; Young, 2005), but recent declines in fertility rates among developed nations have raised concerns among economists and policymakers alike (Bloom et al., 2010; Bloom, Canning and Fink, 2010; Clark et al., 2010; Westley, Choe and Retherford, 2010; Jones, 2022).

2 Data

I utilize the Longitudinal Immigration Database (IMDB) by Statistics Canada, which runs from 1980 through 2021. The IMDB records information about all immigrants, whether refugee or not, upon their arrival to Canada. These data include basic demographic information such as age, education, country of birth, and language, as well as initial destination within Canada. These "landing" data are then linked to tax filings (T1FF) from 1982 through 2019, thus constructing a rich dataset that includes various characteristics about an individual at the time of immigration as well as various behaviors after their arrival in Canada.

I make various adjustments to the universe of immigrants to Canada that arrive between 1980 and 2021. First, I restrict the sample to only women that can be linked across the two component datasets. This necessarily restricts the potential years of immigration, as the data on tax filings is only available through 2019. Second, I restrict the sample to only women between the ages of 20 and 40 to focus on primary years of childbirth.[2] Third, I exclude refugees and immigrants that initially arrive in Quebec. As a semi-autonomous zone within Canada, Quebec separately controls the refugee and immigration processes. Therefore, the refugees arriving in Quebec form a pool completely separate from those arriving in the rest of Canada.

The unit of observation in the dataset is a person-year. Each individual is observed once for every year in which they file taxes and are able to be linked back to the IMDB landing file. Any individual could potentially show up at most 38 times in the IMDB: once every year from 1982 through 2019. Given the sample restrictions I impose, no individual could show up more than 21 times due to the limited age ranges to isolate the highest-fertility period of an individual's life. The average individual appears approximately 10 times.[3] As a result of these sample restrictions, the final dataset includes 20 million observations, representing about 2.2 million different people, for the non-refugee immigrant analysis and 2 million observations, representing about 200 thousand different individuals, for the refugee analysis.

[2]Results are qualitatively similar but slightly deflated if the upper threshold is set at 50 years old instead of 40 years old. This reflects the fact that the vast majority of women above the age of 40 do not give birth in any given year.

[3]This is true for both refugees and non-refugee immigrants regardless of urbanicity.

3 Refugees to Canada

Refugees are individuals forced to flee one country and permanently settle in another. The United Nations Refugee Agency (UNHCR) identifies individuals for resettlement and refers them to Canada, at which point either the Canadian government or private groups sponsor those refugees to travel to Canada for relocation. In addition to granting residence and protection in Canada, refugees benefit from various resettlement services including up to one year of income support, help finding temporary or permanent housing, clothing, food, and information about Canada in general and local services specifically. In some cases, sponsors may provide assistance in one form or another for up to three years.

Canada has a long history of accepting refugees.[4] Approximately 1 million individuals, representing 2.4% of Canada's population and over 10% of all immigrants living in Canada, entered the country as refugees.[5] When refugees are admitted into Canada, they are distributed across the country by the Resettlement Operations Center in Ottawa (ROC-O).[6] ROC-O takes into account various refugee characteristics, such as language and medical needs, and potential destination characteristics, such as existing ethnic communities and the availability of resettlement services. Crucially, the decision is made entirely by the resettlement office and not by refugees, so the final placement in Canada is exogenous from the perspective of the refugees.

[4]For an anecdotal description of Canada's history of refugees, visit https://www.canada.ca/en/immigration-refugees-citizenship/services/refugees/canada-role/timeline.html

[5]Based on calculations from the Public Use Microdata Files from the 2016 Canadian Census.

[6]This applies to all of Canada except Quebec. The Province of Quebec maintains autonomy in selecting refugees separate from the rest of the country.

Table 1 shows the distribution of person-years in which individuals are observed in urban or rural areas, where urban areas are defined as the three largest cities in Canada, broken down by the urbanicity, by the same definition, of their original destination in Canada.[7] The slight majority (53.3%) of refugees are allocated to initial destinations outside the three largest metropolitan areas in Canada. This starkly contrasts with non-refugee immigrants, who explicitly choose their destination within Canada, are shown in Table 2, and for whom the analogous figure is 31.4%. If refugees and non-refugee immigrants had similar preferences on urbanicity upon immigration, then the refugees would presumably immigrate to one of Toronto, Montreal, and Vancouver at much higher rates given the choice of destination.

The different patterns of geographic persistence for refugees and non-refugee immigrants provide additional evidence that initial allocations of refugees do not reflect their preferences for urbanicity. Rural-allocated refugees are observed in urban areas for approximately 29.1% of person-years, suggesting that refugees who are initially assigned rural areas often prefer bigger cities and move there. On the other hand, non-refugee immigrants that initially move to rural areas are much less likely to re-migrate to urban areas; the analogous statistic for non-refugee immigrants is only 21.0%, despite the fact that refugees often have significant incentives to remain in their original destination for some period of time.

This finding is consistent with previous evidence on the destination preferences of refugees. Gure and Hou (2022) finds that refugees strongly prefer large cities over smaller communities and that the majority of refugees are not assigned destinations in accordance with their preferences. This manifests in lower retention of refugees that initially resettle in small communities (Krahn, Derwing and Abu-Laban, 2005; Kaida, Hou and Stick, 2020). Refugees that choose to undertake a secondary migration after initial resettlement in Canada typically choose larger cities as their destination and cite economic reasons for the move (Derwing and Krahn, 2008; Gure and Hou, 2022; Hou, 2007; Hyndman, Schuurman and Fiedler, 2006; Newbold, 1996).

[7] Gure and Hou (2022) provides a more granular breakdown of initial refugee destinations, including how allocations changed between 2000 and 2018.

4 Empirical Strategy

The empirical strategy unfolds in three steps. First, I estimate the following equation on refugee women, for whom their initial placement within Canada is exogenous:

$$P(birth)_{it} = \alpha + \beta Urbanicity_{it} + a_{it} + e_i + k_{it} + l_i + p_i + \delta_t + \varepsilon_{it} \quad (1)$$

where $P(birth)_{it}$ is an indicator equal to 1 if a woman has at least one child born in the past year,[8] $Urbanicity_{it}$ is an indicator equal to 1 if the individual i lives in one of the three largest metropolitan areas of Canada in year t.[9] The right hand side of Equation 1 includes fixed effects for age (a_{it}), maximum education level in the household at arrival in Canada (e_i), number of children above the age of 1 (k_{it}), landing year l_i, place of birth (p_i), and current year δ_t.[10] The error term ε_{it} is clustered at the "destination" Census Metropolitan Area (CMA) level to account for potential correlation in the unobservable component across individuals that are exogenously allocated to a certain location.[11] If refugees stayed in their original destinations within Canada, then β would reflect the causal effect of urbanicity on fertility. However, contemporaneous urbanicity is endogenous in that it reflects individual and household preferences in addition to the exogenous initial place of residence within Canada.

Second, I abstract from contemporaneous urbanicity and instead use the initial destination in Canada as a proxy for urbanicity. I estimate the following equation:

$$P(birth)_{it} = \alpha + \beta Destination_i + a_{it} + e_i + k_{it} + l_i + p_i + \delta_t + \varepsilon_{it} \quad (2)$$

[8] This indicator outcome variable is chosen to abstract from twins, triplets, etc.

[9] The choice of these three largest cities mirrors the definition of large "gateway" cities in Gure and Hou (2022).

[10] Place of birth is defined at the world region level. Table A.1 shows the 26 different world regions, as they are considered in the IMDB. There are six different education groups: none, secondary or less, some postsecondary, bachelor's degree, master or some post-graduate, and doctorate. Age and number of children are not grouped.

[11] There are 131 different Census Metropolitan Areas with positive immigrant counts. The most important correlations are, of course, those for the same individual across time. It is therefore crucial that the level of clustering remain constant for an individual. This is explicitly not the case for various contemporaneous geographic variables, such as CMA or Province (across which individuals may move over time). The choice of clustering variable thus reflects the need to allow for correlations for the same individual across years and is intentionally conservative; results are qualitatively similar if errors are instead clustered at a less conservative (e.g. individual) level.

where $Destination_i$ is instead an indicator equal to 1 if the individual's original destination in Canada was one of the three largest cities.[12] Therefore, the coefficient of interest β in Equation 2 estimates the effect of being initially assigned to an urban area on fertility in year t.

Lastly, I exploit the exogeneity of a refugee's destination and the correlation between original and current location by employing a two-stage least squares (2SLS) specification. I estimate the following equation:

$$P(birth)_{it} = \alpha + \beta \widehat{Urbanicity}_{it} + a_{it} + e_i + k_{it} + l_i + p_i + \delta_t + \varepsilon_{it} \tag{3}$$

where $\widehat{Urbanicity}_{it}$ are the predicted values from the following first-stage equation:

$$Urbanicity_{it} = \gamma + \lambda Destination_i + a_{it} + e_i + k_{it} + l_i + p_i + \delta_t + \epsilon_{it} \tag{4}$$

The urbanicity of an individual is endogenous to their preferences. For refugees who do not choose their initial destination within Canada, this exogenous initial destination serves as an instrument for current urbanicity. This relies on the following assumptions:

1. Destination urbanicity is correlated with urbanicity in a given year.
2. Destination urbanicity does not directly affect fertility in a given year except through the effect on urbanicity in that year.

Given these assumptions, destination urbanicity is a valid instrument for current urbanicity. The first assumption is satisfied as long as destinations are sufficiently "sticky"; this is likely given the support provided to refugees for up to three years to assist resettlement. The second assumption is satisfied as long as the original destination in Canada is as good as random conditional on the various controls included on the right hand side of Equation 4. The relevant F-statistics are 1,032 (Kleibergen-Paap) and 940,000 (Cragg-Donald), suggesting that these instruments are sufficiently strong.

[12] In practice, the original destination must be either Toronto or Vancouver because this specification excludes immigrants that originally migrate to Quebec.

5 Results

5.1 Main Findings

Panel 1 of Table 3 shows the pure differences in fertility rates between rural and urban women. These estimates reflect results from Equations 1-3 without the inclusion of fixed effects for age, number of children education, and place of birth; estimated coefficients therefore represent differences in average birth rate across urbanicity. Column 1 shows the results from non-refugee immigrants, included here to provide context on rural-urban differences in birth rates for individuals that explicitly select their destination within Canada. Column 2 provides results from Equation 1 using only refugees, who are exogenously allocated to initial destinations within Canada. Column 3 provides results from Equation 2 using only refugees, thus estimating the effect of being initially allocated to an urban area on fertility in any given year (regardless of urbanicity in that year). Column 4 shows the results from the two-stage least squares specification, in which original destination urbanicity within Canada serves as an instrument for current urbanicity. All four columns, using different populations and specifications, demonstrate significant rural-urban fertility differences.

The coefficients in Column 1 and Column 2 document the lower birth rates in urban areas relative to rural areas of 0.99 and 1.48 fewer births per 100 women for non-refugee immigrants and refugees, respectively. The magnitudes of the estimated coefficients are lower using destination urbanicity as either a proxy (49.3% lower) or an instrument (12.8% lower) for contemporaneous urbanicity. However, the effect of urbanicity on fertility is strong and statistically significant even exploiting the exogeneity of initial refugee destinations.

Panel 2 of Table 3 provides estimates for the same coefficients in Equations 1-3 with the inclusion of fixed effects for age, number of children, education, place of birth, year, and landing year. These results thus account for potential selection on observable characteristics, either in the refugee allocation process or in the choice of location for any given year. For all four columns, estimated coefficients are qualitatively similar across Panel 1 and Panel 2; although the magnitude of the coefficients is slightly reduced relative to Panel 1, they remain strongly negative and statistically significant.[13] Surprisingly,

[13] Tables A.2 through A.5 show results from gradually including additional fixed effects. Results are qualitatively similar to Panel 1 and Panel 2 regardless of the exact specification.

this suggests that selection on these six observable characteristics (age, education level, number of existing children, place of birth, year, and landing year) contributes, at most, moderately to rural-urban fertility differences.

The coefficient in Column 2 shows the rural-urban fertility difference, after controlling for those six observable characteristics, for refugees based on their contemporaneous urbanicity. The estimated coefficient of interest from Equation 2 shows the causal effect of being assigned to an urban area is only 46.2% of the same from Equation 1. This suggests that, even accounting for various observable characteristics, the effect of being assigned to an urban area contributes approximately half of the observed rural-urban fertility gap among refugees in Canada. Furthermore, the results from the two-stage least squares specification suggest that the true causal estimate of urbanicity on fertility is a 0.85 percentage point decrease in the annual probability of giving birth. This coefficient is statistically significant at the 1% level and represents a 9.1% decrease on the baseline level of births per refugee woman in rural areas for a given year.[14]

5.2 Robustness Checks

I conduct various robustness checks considering alternative samples of refugees and augmented empirical strategies. Although none of the refugees considered here can explicitly choose their destinations within Canada, this exogeneity is most prevalent for government-assisted refugees (GAR's).[15] Table A.6 replicates Table 3 using only government-assisted refugees, and the results are quantitatively similar; estimated coefficients and standard errors are nearly identical to those from the primary specification.

The primary sample excludes refugees who initially arrive in Quebec because the province, as a semi-autonomous zone within Canada, retains significant control over their own immigration and refugee processes. However, the results are quantitatively similar when immigrants to Quebec are included. Table A.7 replicates Table 3 using the expanded sample; although the estimated coefficient for non-refugee immigrants is much lower relative to Table 3, the coefficients for refugees (Columns 2-4) are nearly identical. This is unsurprising given that NRI's and refugees to Quebec comprise, at most,

[14]This estimate is in line with the raw rural-urban gap in births per woman for non-refugee immigrants, as is presented in Panel 1 Column 1: 11.6%.

[15]GAR's are approximately 53.7% of the refugee sample considered.

20% of the relevant samples.

Although the IMDB includes the universe of immigrants to Canada since 1980, I only have yearly data on individuals that file their taxes in a given year. This is crucial given that the outcome of interest is one-year fertility (as opposed to total fertility). It is theoretically possible that this potential for attrition may bias estimated coefficients if the probability of filing taxes is correlated with fertility. I thus exclude all immigrants that fail to file taxes in any year between their first and last filing; results are presented in Table A.8 and are quantitatively similar to those in Table 3.

The primary specification considers only the three "gateway" cities of Toronto, Montreal, and Vancouver urban; those are by far the largest metropolitan areas in Canada and are thus categorized here as such. However, Canada has many other medium-sized cities that could also be considered urban by a less restrictive definition. Table A.9 presents results from an alternative specification in which Calgary, Edmonton, and Ottawa are also considered in the urban group rather than the non-urban group. Estimated coefficients are somewhat attenuated towards zero, although they remain strongly negative and statistically significant, especially in Panel 2.

Lastly, Tables A.10 and A.11 present results from Equation 1 with alternative fixed effects for place of birth and education, respectively. Whereas the primary specification categorizes place of birth by world region, Table A.10 instead uses the country of birth. Whereas the primary specification assigns education fixed effects based on the maximum education level for any household member upon arrival, Table A.11 instead uses the realized education of the refugee woman in question upon arrival. In both cases, the results are quantitatively similar.

6 Conclusion

These findings suggest that causal factors are primarily responsible for rural-urban differences in fertility. The gap in birth rates persists among refugees, for whom placement within Canada is exogenous, via a naive OLS analysis, analysis using original destination urbanicity as a proxy for current-year urbanicity, and analysis using original destination in Canada as an instrument for current-year urbanicity. The latter estimated coefficient reflects the direct causal impacts of urbanicity on fertility. This result is statistically significant at the 1% level and economically similar to the OLS estimates for both refugees and non-refugee immigrants, thus suggesting that contextual factors, rather than compositional factors, not just contribute to but in fact drive observed rural-urban differences in birth rates.

The results also indicate that "socialization" is not the driving causal mechanism through which urban areas lead to lower fertility rates. Due to data limitations, this project specifically analyzes women that arrived in Canada after the age of 18. The fact that rural-urban differences are similar between refugees and non-refugee immigrants subject to this age restriction suggests that time spent in rural or urban areas prior to adulthood does not affect eventual fertility. This mechanism deserves attention from future research, which may exploit time spent in rural or urban areas by children who arrive in Canada as refugees (and are thus exogenously allocated to different parts of the county).

There are various limitations with this analysis. First, the refugee placement process is exogenous rather than random. Specifically, private sponsors and the government may still take into account various observable characteristics in the allocation process. For example, refugee households with children may be more likely to be placed in rural areas. Therefore, the groups that eventually move to rural and urban areas are not necessarily ex-ante identical. Although the explicit identification assumption required here is only that refugee allocation is random conditional on the observable characteristics included on the right hand side of Equation 4, this still represents a departure from the ideal natural experiment

Second, this analysis focuses on refugees moving to a new country fleeing persecution. These individuals, by construction, are adjusting to a new location and a new culture. While this is helpful for the purposes of their exogenous placement within Canada, the situation is extremely stylized and does not reflect the circumstances under which rural-urban differences in fertility are typically observed. Although this exogenous variation is necessary for identification in the context of this paper, the specialized situation of these refugees is not typical and may lack external validity. Future analysis might exploit quasi-experimental variation in urbanicity in a different context in order to add to the literature on this topic and further contribute to our understanding of rural-urban fertility differences.

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Table 1: Geographic Persistence - Refugees

		Destination	
		Rural	Urban
Current	Rural	4,410,000 (37.8%)	840,000 (7.2%)
	Urban	1,810,000 (15.5%)	4,600,000 (39.5%)
		53.3%	46.7%

The top-left cell shows the number of person-years in which individuals that were originally allocated to rural areas upon arrival in Canada are observed in rural areas. The bottom-left cell shows the number of person-years in which individuals that were originally allocated to rural areas upon arrival in Canada are observed in urban areas, highlighting the prevalence of rural-allocated refugees eventually moving to urban areas. The top-right cell shows the number of person-years in which individuals that were originally allocated to urban areas upon arrival in Canada are observed in rural areas. The bottom-right cell shows the number of person-years in which individuals that were originally allocated to urban areas upon arrival in Canada are observed in urban areas. Parentheses show the percentage of total person-years represented by a given cell. Only refugees are included in this table.

Table 2: Geographic Persistence - NRI

		Destination	
		Rural	Urban
Current	Rural	22,220,000 (24.8%)	7,540,000 (8.4%)
	Urban	5,890,000 (6.6%)	53,950,000 (60.2%)
		31.4%	68.6%

The top-left cell shows the number of person-years in which individuals that were originally allocated to rural areas upon arrival in Canada are observed in rural areas. The bottom-left cell shows the number of person-years in which individuals that were originally allocated to rural areas upon arrival in Canada are observed in urban areas, highlighting the prevalence of rural-allocated refugees eventually moving to urban areas. The top-right cell shows the number of person-years in which individuals that were originally allocated to urban areas upon arrival in Canada are observed in rural areas. The bottom-right cell shows the number of person-years in which individuals that were originally allocated to urban areas upon arrival in Canada are observed in urban areas. Parentheses show the percentage of total person-years represented by a given cell. Only non-refugee immigrants are included in this table.

Table 3: Rural-Urban Fertility Differences

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0099**	-0.0148***	-0.0075***	-0.0129***
	(0.0039)	(0.0013)	(0.0009)	(0.0015)
CD F-stat				960,000
RK F-stat				906
Panel 2				
Urban	-0.0088***	-0.0104***	-0.0048***	-0.0085***
	(0.0024)	(0.0011)	(0.0008)	(0.0014)
Fixed Effects	X	X	X	X
CD F-stat				940,000
RK F-stat				1,032
Rural Mean	0.0857	0.0935	0.0891	
N	19,230,000	2,010,000	2,010,000	2,010,000
People	2,180,000	190,000	190,000	190,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for refugees. Column 3 presents the difference in fertility rates between refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity. The F-statistic in Panel 2 Column 4 replicates the F-statistic from Column 5 of Table A.5; updated F-statistics for additional specifications have not yet been released from the Research Data Centre.

A Additional Tables

Table A.1: SCCAI World Areas

World Area	Continent	SCCAI Code
North America	Americas	11
Central America	Americas	12
Caribbean and Bermuda	Americas	13
South America	Americas	14
Americas, not elsewhere stated	Americas	19
Western Europe	Europe	21
Eastern Europe	Europe	22
Northern Europe	Europe	23
Southern Europe	Europe	24
Europe, not elsewhere stated	Europe	29
Western Africa	Africa	31
Eastern Africa	Africa	32
Northern Africa	Africa	33
Middle Africa	Africa	34
Southern Africa	Africa	35
Africa, not elsewhere stated	Africa	39
West Central Asia and the Middle East	Asia	41
Eastern Asia	Asia	42
Southeast Asia	Asia	43
Southern Asia	Asia	44
Asia, not elsewhere stated	Asia	49
Oceania	Oceania	51
Oceania, not elsewhere stated	Oceania	59
Antarctica and Adjacent Islands	Antarctica	69
Antarctica, not elsewhere stated	Antarctica	69
Not applicable	N/A	88

Table A.2: Non-Refugee Immigrants - Current Urbanicity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Urban	-0.0099**	-0.0093**	-0.0099**	-0.0105***	-0.0092***	-0.0095***	-0.0088***
	(0.0039)	(0.0036)	(0.0039)	(0.0037)	(0.0025)	(0.0024)	(0.0024)
Rural Mean	0.0857	0.0857	0.0857	0.0857	0.0857	0.0857	0.0857
Age Fixed Effects		X	X	X	X	X	X
# Kids Fixed Effects			X	X	X	X	X
Educ Fixed Effects				X	X	X	X
POB Fixed Effects					X	X	X
Year Fixed Effects						X	X
Landing Year Fixed Effects							X
N	19,230,000	19,230,000	19,230,000	19,230,000	19,230,000	19,230,000	19,230,000
People	2,180,000	2,180,000	2,180,000	2,180,000	2,180,000	2,180,000	2,180,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Column 1 shows the raw rural-urban fertility differences among non-refugee immigrants, thus replicating Column 1 of Table 2 Panel 1. Columns 2 through 5 show the same results with the gradual inclusion of additional fixed effects. Column 5 replicates Column 1 of Table 2 Panel 2.

Table A.3: Refugees - Current Urbanicity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Urban	-0.0148***	-0.0143***	-0.0160***	-0.0157***	-0.0146***	-0.0134***	-0.0104***
	(0.0013)	(0.0012)	(0.0011)	(0.0011)	(0.0010)	(0.0008)	(0.0011)
Rural Mean	0.0935	0.0935	0.0935	0.0935	0.0935	0.0935	0.0935
Age Fixed Effects		X	X	X	X	X	X
# Kids Fixed Effects			X	X	X	X	X
Educ Fixed Effects				X	X	X	X
POB Fixed Effects					X	X	X
Year Fixed Effects						X	X
Landing Year Fixed Effects							X
N	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000
People	190,000	190,000	190,000	190,000	190,000	190,000	190,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the ”Destination CMA” level in parentheses.

Column 1 shows the raw rural-urban fertility differences among non-refugee immigrants, thus replicating Column 2 of Table 2 Panel 1. Columns 2 through 5 show the same results with the gradual inclusion of additional fixed effects. Column 5 replicates Column 2 of Table 2 Panel 2.

Table A.4: Refugees - Destination Urbanicity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Urban	-0.0075***	-0.0070***	-0.0079***	-0.0070***	-0.0068***	-0.0060***	-0.0048***
	(0.0009)	(0.0009)	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0008)
Rural Mean	0.0891	0.0891	0.0891	0.0891	0.0891	0.0891	0.0891
Age Fixed Effects		X	X	X	X	X	X
# Kids Fixed Effects			X	X	X	X	X
Educ Fixed Effects				X	X	X	X
POB Fixed Effects					X	X	X
Year Fixed Effects						X	X
Landing Year Fixed Effects							X
N	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000
People	190,000	190,000	190,000	190,000	190,000	190,000	190,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Column 1 shows the raw rural-urban fertility differences among non-refugee immigrants, thus replicating Column 3 of Table 2 Panel 1. Columns 2 through 5 show the same results with the gradual inclusion of additional fixed effects. Column 5 replicates Column 3 of Table 2 Panel 2.

Table A.5: Refugees - 2SLS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Urban	-0.0129***	-0.0121***	-0.0137***	-0.0122***	-0.0118***	-0.0105***	-0.0085***
	(0.0015)	(0.0014)	(0.0013)	(0.0012)	(0.0013)	(0.0014)	(0.0014)
Age Fixed Effects		X	X	X	X	X	X
# Kids Fixed Effects			X	X	X	X	X
Educ Fixed Effects				X	X	X	X
POB Fixed Effects					X	X	X
Year Fixed Effects						X	X
Landing Year Fixed Effects							X
CD F-stat	960,000	960,000	960,000	950,000	940,000	940,000	940,000
RK F-stat	906	901	885	846	1,032	1,032	1,032
N	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000	2,010,000
People	190,000	190,000	190,000	190,000	190,000	190,000	190,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Column 1 shows the raw rural-urban fertility differences among non-refugee immigrants, thus replicating Column 4 of Table 2 Panel 1. Columns 2 through 5 show the same results with the gradual inclusion of additional fixed effects. Column 5 replicates Column 4 of Table 2 Panel 2.

F-statistics in Column 6 and Column 7 simply replicate the F-statistics from Column 5; updated F-statistics for additional specifications have not yet been released from the Research Data Centre.

Table A.6: Government-Assisted Refugees

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0099**	-0.0170***	-0.0076***	-0.0132***
	(0.0025)	(0.0011)	(0.0008)	(0.0013)
Panel 2				
Urban	-0.0088***	-0.0114***	-0.0049***	-0.0087***
	(0.0024)	(0.0011)	(0.0014)	(0.0022)
Fixed Effects	X	X	X	X
N	19,230,000	1,080,000	1,080,000	1,080,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.

Table A.7: Including Quebec-destined Immigrants

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0063*	-0.0137***	-0.0064***	-0.0112***
	(0.0037)	(0.0014)	(0.0013)	(0.0022)
Panel 2				
Urban	-0.0067***	-0.0102***	-0.0046***	-0.0083***
	(0.0022)	(0.0012)	(0.0008)	(0.0015)
Fixed Effects	X	X	X	X
N	23,130,000	2,460,000	2,460,000	2,460,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.

Table A.8: Excluding Refugees with Tax-filing Gaps

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0100**	-0.0150***	-0.0074***	-0.0124***
	(0.0041)	(0.0014)	(0.0011)	(0.0016)
Panel 2				
Urban	-0.0091***	-0.0106***	-0.0051***	-0.0087***
	(0.0025)	(0.0011)	(0.0008)	(0.0014)
Fixed Effects	X	X	X	X
N	16,390,000	1,680,000	1,680,000	1,680,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the ”Destination CMA” level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.

Table A.9: Lower Urbanicity Threshold

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0066**	-0.0075***	-0.0041*	-0.0087*
	(0.0033)	(0.0016)	(0.0023)	(0.0049)
Panel 2				
Urban	-0.0062***	-0.0073***	-0.0035**	-0.0073**
	(0.0024)	(0.0017)	(0.0014)	(0.0030)
Fixed Effects	X	X	X	X
N	19,230,000	2,010,000	2,010,000	2,010,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.

Table A.10: Country-of-Birth Fixed Effects

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0099**	-0.0148***	-0.0075***	-0.0129***
	(0.0039)	(0.0013)	(0.0009)	(0.0015)
Panel 2				
Urban	-0.0072***	-0.0106***	-0.0052***	-0.0092***
	(0.0020)	(0.0014)	(0.0008)	(0.0016)
Fixed Effects	X	X	X	X
N	19,230,000	2,010,000	2,010,000	2,010,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.

Table A.11: Individual Education Fixed Effects

Panel 1	NRI	OLS	Destination	2SLS
Urban	-0.0099**	-0.0148***	-0.0075***	-0.0129***
	(0.0039)	(0.0013)	(0.0009)	(0.0015)
Panel 2				
Urban	-0.0090***	-0.0106***	-0.0050***	-0.0088***
	(0.0025)	(0.0012)	(0.0008)	(0.0016)
Fixed Effects	X	X	X	X
N	19,230,000	2,010,000	2,010,000	2,010,000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the "Destination CMA" level in parentheses.

Panel 1: Column 1 presents the the difference between fertility rates in rural and urban areas for non-refugee immigrants. Column 2 presents the difference for government-assisted refugees. Column 3 presents the difference in fertility rates between government-assisted refugees that were initially allocated to rural areas and those that were initially allocated to urban areas, regardless of current location. Column 4 presents the estimated coefficient of interest from Equation 3 without the inclusion of fixed effects.

Panel 2: Column 1 presents the estimated coefficient β from Equation 1 estimated using only non-refugee immigrants. Column 2 presents the same using only government-assisted refugees. Column 3 presents the estimated coefficient β from Equation 2, in which current urbanicity is replaced by a refugee's original destination in Canada. Therefore, this coefficient represents the effect, in any given year, of being initially assigned to an urban destination.

Column 4 presents the estimated coefficient from Equation 3, the second stage of a 2SLS specification in which current urbanicity is instrumented by destination urbanicity.